
THE STATISTICS OF NATURAL EXPERIENCE

(& ITS CONSEQUENCES FOR HOW WE THINK ABOUT LEARNING AND INTELLIGENCE)

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LECTURES BY LINDA B. SMITH

Dota Tianai Dong^{*1}, Jing Li^{*2}, Tobias Thomas^{*3}, and Linda B. Smith⁴

¹Max Planck Institute for Psycholinguistics, Nijmegen, the Netherlands

²Icahn School of Medicine at Mount Sinai, New York City, NY, USA

³Technical University Darmstadt, Darmstadt, Germany

⁴Indiana University, Bloomington, IN, USA

^{*}These authors contributed equally to this work. Author order is alphabetical.

ABSTRACT

These lecture notes cover the content from Linda B. Smith at the 2024 edition of the Analytical Connectionism Summer School¹. Her lecture focused on the active and dynamic nature of infant learning, especially how the statistics of the environment shape the learning process and vice versa. In the beginning she lines out a brief history of the field and how the broad availability of data capturing devices like head-mounted cameras or eye-trackers have made the line of research that investigates the interplay of input statistics and learning possible in the first place. She follows that up by discussing the natural visual curriculum of early infancy and how their attention shifts from faces gradually to more complex scenes, like hand-object interactions. Next, Linda discusses key coherence statistics and argues that their actively shaped regularities are an important foundation of human intelligence. Last, she discusses the key social learning scenario between a child and its parent, highlighting how joint attention and active input selection drive learning for the kid. Throughout the lectures, Linda presents evidence that using infants' natural learning curricula can improve the performance and/or learning speed of artificial intelligence systems, thereby linking research on human intelligence and learning with research on artificial systems

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1 Lecture 1: Statistics shape the interaction between behavior and learning

1.1 Introduction and preliminaries

Human infants are remarkably efficient learners and statistics of their self-generated experiences play a crucial role in enabling that efficiency. Thus, studying their development and how they acquire a vast range of knowledge and skills from their naturalistic experience can not only help us better understand human nature but may help us to develop better and more efficient artificial intelligence systems. Some work in machine learning has already picked up some aspects of the special curriculum that is early infant experience and used such data for learning, showing that even artificial learning systems in a wide variety of tasks profit from a structured curriculum [1–5]. Yet, that still leaves the questions of why it is so efficient and how a human baby acquire these inputs unanswered. The changes in the brain that result from this learning process happen on very short timescales of milliseconds to seconds and during every interaction with the environment. To understand these changes, it is important to analyze the statistical properties of the sensory inputs. There are two properties of those statistics that have been shown to be key factors driving human learning, while barely being considered when training AI systems. First, all natural inputs available to human infants are inherently constrained by the physics of space and time, arising from their embodied and sequential sensory experience. In contrast, data points in most machine-learning datasets are largely independent of one another, often lacking the egocentric and temporal structure that characterizes infant learning. Second, the learner, in this case, a human baby, actively selects, creates and actively controls their inputs to make learning and development more efficient. Learning algorithms can exploit both of these properties to be more efficient [3, 5]. Most scientific fields that study some form of intelligence, from neuroscientists to behavioral psychologists to computational modelers, focus on scenarios where the input is fixed and try to understand how this given input drives behavior or some other internal measurements. Although this notion is changing and especially in recent years with, for example, the subfield of active learning in AI, or the push towards naturalistic tasks in experimental psychology and cognitive science, more research focuses on non-stationary environments. Importantly, none of these processes happen independently, but there exists a closed loop between development, behavior, brain activity, and the statistics of the inputs [6]. Figure 1 shows the idea of how these concepts interact with each other.

Starting approximately 20 years ago, made possible through technological advancements, researchers started quantifying the statistics of natural inputs of human children using a mixture of head-mounted cameras, (head-mounted) eye-trackers, motion-tracking systems, and other methods [7–9]. There is one important thing to note for the many studies that have used head-mounted cameras to record a head-centered field of view of the subject wearing the device. Due to the properties of the human visual system, which only has high acuity at the fixation point, without eye-tracking, one can not be sure at which point in the visual field a subject is looking. Yet, the literature provides ample evidence that if people, independent of age, are allowed to move their head and body freely, most of their gaze will be around the center of the head-centered field of view [10, 11]. Thus it is possible to draw conclusions about the point of visual attention by using head-mounted cameras only instead of additional eye-trackers.

The following will outline three examples providing evidence for the claims made thus far and vividly show how infants shape their own inputs to learn efficiently.

1.2 Example 1: Biased looking at the scale of daily life

Prior studies have shown that young infants, not older than 5 months, prefer to view simple visual stimuli [12]. These are characterized by low spatial frequencies as well as high contrasts, or in other words images with large areas of similar content and sharp edges. This specific experience is widely viewed as the one that is crucial in tuning the edge receptive fields in early visual processing and missing it has been shown to lead to long-term visual impairments [13, 14]. Yet, what is missing is a quantitative measurement of the visual input during those crucial times of development to answer the question how the known preference for simple stimuli influences infant behavior.

To answer this question, Anderson *et al.* [15] have collected head-centered field-of-view videos from 10 infants between 3 and 13 weeks old, as well as from 10 adults between 33 and 71 years old. All subjects wore the head-mounted video cameras multiple hours daily, leading to a representative representation of their natural visual inputs. Both groups have contributed over 30 hours of videos, which were downsampled to 50000 images, equally split between infants and adults. The final 25000 images per group corresponded to a frequency of 0.2 and 0.25 images per second for adults and infants respectively.

The resulting images were then analyzed for edge simplicity and edge visibility. Edge simplicity was measured by edge sparsity (how many pixels in the image belong to an edge) and orientation consistency (how many edges belong to the most frequent orientation). Edge visibility was measured by the root means squared (RMS) contrast over the whole image and the slope of the amplitude spectrum (the amount of log contrast as a function of log spatial frequency).

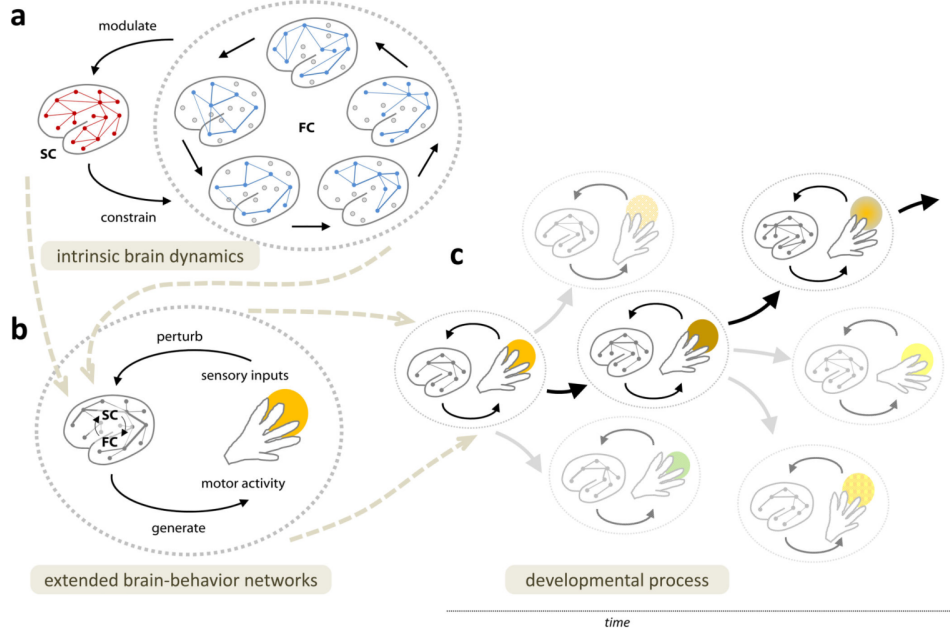


Figure 1: Brain, body, and behavior dynamically interact over time to shape the developmental process. **a** Functional connectivity (FC) networks evolve quickly but are constrained by structural connectivity (SC), which evolves over longer timescales. **b** Movement is the bridge of the brain with the environment. Motor neurons generate behavior, while behavior, on the other hand, shapes sensory inputs that modulate SC and FC. **c** Both parts of the brain-behavior dynamic undergo significant changes over the course of development. From Byrge *et al.* [6].

Both measures, simplicity and visibility, were reduced from two features to a single scalar value by performing an independent principal component analysis (PCA) and only considering the respective first dimension. Afterward, images were clustered in this 2D space of edge simplicity and visibility into three categories: *simple*, *complex*, and *texture*. Figure 11 shows all images in the simplicity-visibility space as well as the category they got clustered into. The analysis shows that *texture* is the most infrequent category for both infants and adults, with less than 5% of images falling into the category for both groups. While the rest of the adult images fall nearly equally into the *simple* and *complex* categories (51.0% and 44.9% respectively), the remaining infant images fall predominantly into the *simple* category (72.7%), while significantly fewer are categorized as *complex* (25.0%). The cluster analysis confirms that infants also have a preference for simpler visual stimuli in natural environments.

While the question of how infants achieve to select their visual inputs is still not fully answered, they use head and eye movements to orient themselves toward simpler edge patterns. The study shows that human infants are in fact biased learners and actively select their inputs to maximize learning efficiency and crucially do so even in complex natural environments.

1.3 Example 2: Optimizing the input

Given a particular task goal, how should one allocate attention to achieve this goal? This is one of the fundamental questions of vision science. Understanding the properties of attentive periods that lead to effective learning would be a step in answering this question. Saliency research has pointed out features of visual objects that naturally draw attention, like size, color, and centeredness in the visual field [16–18].

One common paradigm to study this question is object name learning. In these experiments, infants play together with one of their parents and three (this number can vary) different toys [9, 19, 20]. Parents are told the names of the toys beforehand but do not know that children will later get tested on the names. Most of these experiments also make use of head-mounted cameras, eye trackers, and motion trackers for the hands.

The first result that all of these studies report is that the pure number of naming events is not a predictor of whether the child learns the object’s name or not. Instead, attentional properties like centeredness and the distance of the named object to distractors are much better predictors [19, 21]. Another factor that helps object name learning is the toddler holding the object in their own hands. It stabilized not only the head but also the whole body around it, leading to very

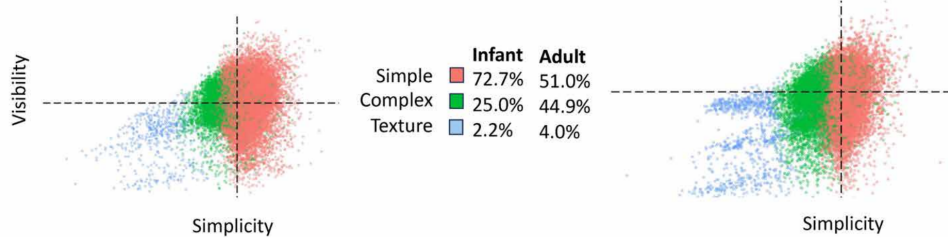


Figure 2: Distribution of infant (left) and adult (right) images in the simplicity-visibility space. The color of each dot indicates the cluster to which they were assigned in the cluster analysis. The table in the middle shows the proportions of the three clusters for both adults and infants. From Anderson *et al.* [15]

stable visual inputs and thus making learning easier. To maximize the learning probability, these properties have to hold a significant amount of time, a minimum of 3, better 5 seconds, around the moment of naming the object. Figure 3 visualizes these results for object sizes by comparing the size of targets and distractors for naming periods that lead to learning and those that did not.

The data also shows that to enable toddler learning, the named object needs to be greater in size than all distractors, and this time window needs to start before the naming event itself and hold for a prolonged time afterward.

These elongated times of attention on a single object, called *sustained attention*, have long been researched and have been shown to be a good predictor for performance in a variety of tasks as well as for long-term development and self-regulatory behavior during adult life [22–24]. To study these sustained attention periods, Mendez *et al.* have recently conducted a study with 12-month-old toddlers where they got to play in 45-second-long exploration periods with three different toys. Infants’ gazes were also recorded using head-mounted eye trackers.

First analysis showed that in 71% of the frames, the infants’ gaze was on the largest object in the scene and 19% of the attention periods were above the threshold of 3 seconds and thus counted as sustained attention periods. Analyzing the sustained attention periods can help to reveal the causal structure for the fact that most gaze falls onto the largest object in the scene. Figure 4 shows the relative (a) and absolute (b) size of the attended object around the time of the look onset and offset. The data clearly highlights that subjects actively increase objects’ absolute and relative size briefly before attending them. When their gaze leaves the object, its size also drastically decreases again. This provides evidence for the fact that it is not big objects that attract attention, but attention leads to the toddler making head and body movements to increase the size of the attended object.

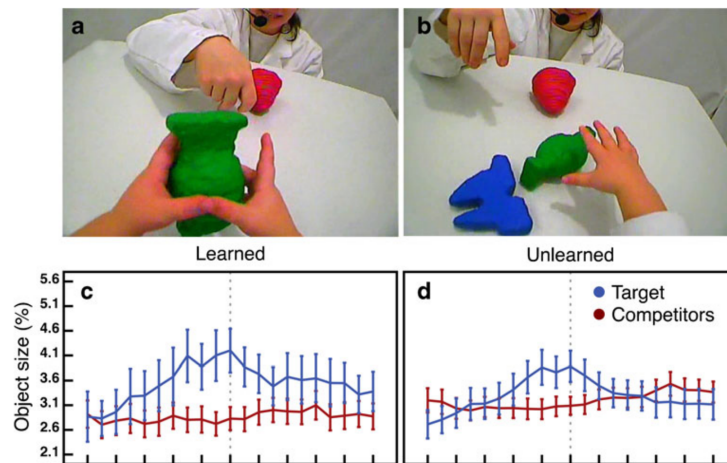


Figure 3: Differences around naming events that influence object name learning. **a,b** Two example scenes from the toddlers’ field of view camera at the time point where the parent named the object. One scene (a) where the child did learn the name of the relevant object (green) and one (b) where it did not learn the name (blue). **c-d** Object size, in percentage of the visual field, over time for learned naming events (c) and unlearned naming events (d). Both graphs show the 20 second window ($\pm 10s$) around the naming event itself. From Pereira *et al.* [19]

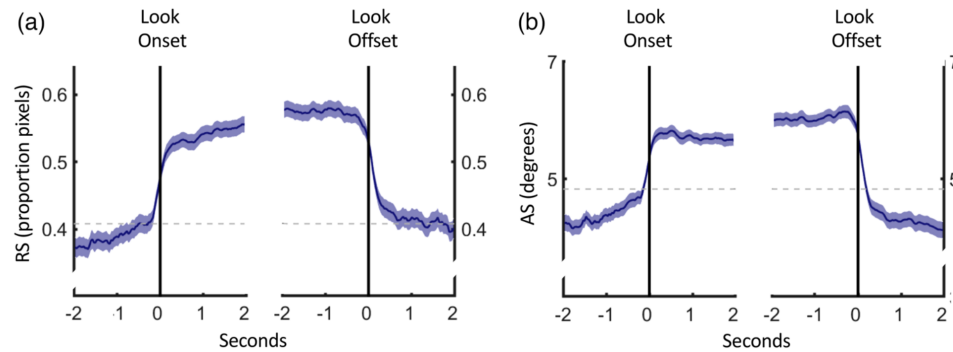


Figure 4: Visual size changes of objects around the onset and offset of sustained attention periods. **a,b** Mean relative size (RS) (a) and absolute size (AS) (b) of the focused object for the ± 2 s time interval around the onset (left) and offset (right) of all sustained attention (longer than 3s) periods. All graphs show the mean curve in bold and the standard error of the mean as a shaded area. From Mendez *et al.* [8]

This study again provides evidence for the fact that human infants are aware of the conditions that enable themselves to maximize learning performance, in this case large and centered positioning of the relevant object. Furthermore they actively use their own motion to control the visual input to fit into the previously described conditions.

1.4 Example 3: Object views and interaction

Not only object recognition but most interaction with objects in our everyday life depends on the views of said object [25]. From an adult standpoint, most objects have a handful of natural positions and axes, which we rotate around when interacting with them. These are their upright position, and the axes are the symmetry axes and the axis of major extent. Young children, on the other hand, might not know what the natural upright position of an object is or understand the concept of symmetry at all. That poses the question: when infants interact with objects, do they use similar views as adults would?

To answer this, Pereira *et al.* conducted a study where they let children between the ages of 12 and 36 months old interact with a range of objects from known to nonsense [26]. The child's field of view was captured using head-mounted cameras. They were used to determine the current orientation of the object in space.

Figure 5 shows the dwell times averaged over all objects of the infants split into four age groups. This figure shows the summary of known and nonsense objects because there was no difference between the two categories. There are two important trends to notice in the data. First, children get more structured over time. While 12-month-olds are still quite messy and use a lot of different orientations, the older the infants get, the more dwell time they spend around the planar views of the object. Second, even though the different age groups are on different levels of consistency, the overall structure, where most of the dwell time is spent, and which views are completely avoided, is the same.

A straightforward hypothesis where these biases are coming from is that all humans, independent of age, hold the objects in their hands when manipulating them. So maybe the planar views correspond to practical hand positions, making them attractive independent of age. Yet, a much more compelling argument is that everyone wants to interact with these objects and is thus seeking the views of the objects that maximize their information. There is quite a long history of vision science research providing evidence that these planar views provide exactly that [27–30]. They provide a certain frame of reference and a stable rate of change, something that Marr & Nishikara have called non-accidental properties and often also some forms of symmetry [27].

This is another study providing evidence that infants are from an early stage intrinsically motivated to learn to interact with objects in the world. Additionally, they already start out with a good model that very quickly matures and stabilizes to choose their own actions and manipulate the objects around them to maximize information and thus make learning more quicker and more efficient.

1.5 Conclusion

We started this lecture with the hypothesis that young infants' interaction between behavior and learning are not only passively using their own sensory inputs, but also actively shape these inputs to optimize for learning and behavior. Figure 1 visualizes this idea of how brain, body, and behavior interact with each other over different timescales. For the

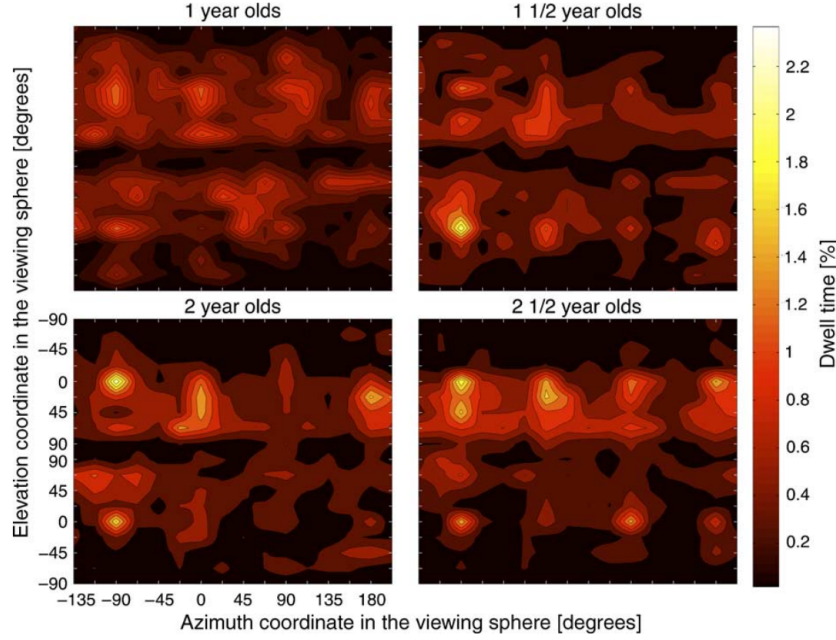


Figure 5: Dwell time map for all four age groups in the experiment. For all plots, the upper half corresponds to objects in an upright rotation and the lower half in reverse. Elevation values were ordered such that each object is symmetrical around the map’s horizontal midline. For visual examples of rotated objects, see the original paper [26]. From Pereira *et al.* [26]

rest, we have discussed three examples that provide not only evidence for this hypothesis but also show how valuable understanding environmental statistics is to understanding how infants behave and learn. The first example showed that the result from laboratory studies that children of a young age prefer simple visual stimuli also transfers to their everyday life, where children make use of their head and body movement to select simple scenes. The second example showed how important stable visual inputs are for learning and how, as a consequence of that, children actively use sustained periods of attention to learn a variety of properties, e.g., the names of objects. The third example showed that children of all ages make use of symmetries and planar views of objects to maximize the information that they gain about the object. These behaviors can barely be taught because they start at a very young age and happen independently of the type of interaction with the object.

2 Lecture 2: A curriculum

2.1 Introduction and preliminaries

Learning—defined as the ability to make adaptive, intelligent changes based on experience—represents both a fundamental aspect of human cognitive development and a long-pursued objective in artificial intelligence research. A prevalent (mis)conception in machine learning research is that effective learning emerges primarily through passive exposure to massive datasets, as exemplified by training large models on extensive collections like ImageNet. In contrast, children demonstrate remarkable capabilities in acquiring knowledge rapidly and effectively from limited examples, showing both impressive generalization across contexts and precise discrimination between similar but distinct cases. What appears essential for human learning—and what likely enables our extraordinary learning abilities—is a structured, curriculum that builds knowledge progressively through carefully sequenced experiences. In machine learning, analogous ideas are formalized in Bayesian active learning approaches [31–33], which provide algorithms for optimally selecting data points to reduce training time. Such methods may help explain certain elements of infants’ curriculum-driven learning. In this context, Linda argues that the data for visual learning changes significantly with each advancement in motor abilities, creating a natural curriculum for visual learning that’s ordered by the developing sensory-motor capabilities of the learner [35, 36]. The “data” provides a crucial element for learning as it not only defines the task to be solved but also contains the essential knowledge to be acquired. In the following section, Linda examines what we know about the infant’s learning environment, how information is structured in the natural world to facilitate this remarkable developmental process, and its implications for machine learning.

To investigate this question empirically, Linda and her team have conducted an extensive research program utilizing head-mounted cameras with eye trackers to capture the detailed visual experiences of infants during everyday object interactions over time. This methodology precisely documents how infants’ bodies and behaviors select and structure their visual experiences, consisting of datasets from multisensory environments (Mutisensory), lab settings (Toyroom), and home environments (Homeview & Indiaview). This decade-long research initiative follows infants from 1 month to 24 months of age, compiling over 1,000 hours of head-camera video footage, which has enabled many studies that train AI models using the realistic amount of data [3, 37].

2.2 From the dense with faces to dense with hands

In this section, Linda presents several findings from the Homeview and Indiaview projects, which comprise an extensive corpus of developmentally indexed egocentric scenes captured from 101 infants in the United States and 50 infants in India. She begins by highlighting a pivotal study conducted in her lab that examines the prevalence of faces in infant-perspective scenes and systematically documents how these visual experiences evolve throughout the first year of life. For more comprehensive details on this work, please see the original paper [34].

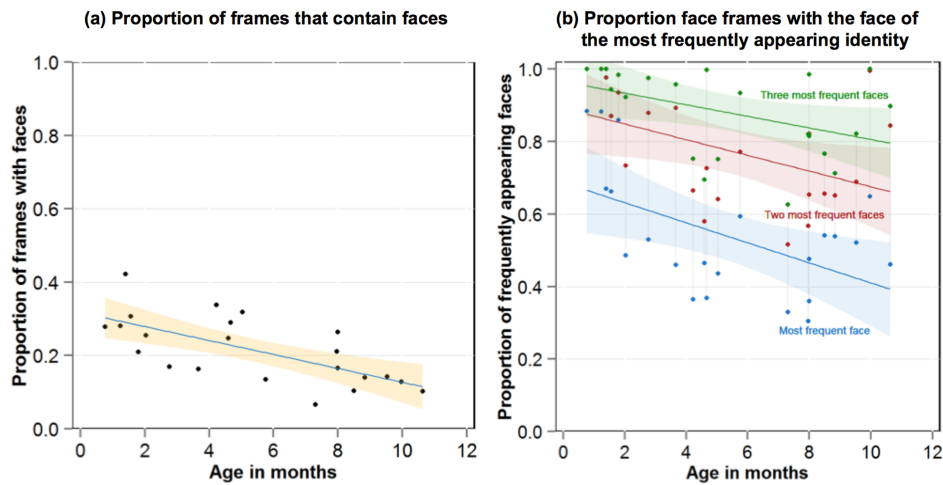


Figure 6: (a) tracks how often babies see faces as they grow older, with the solid line showing the average trend and the shaded area showing the statistical range. (b) shows which faces babies see most often - the blue line tracks their most common face (like mom), the red line shows their two most common faces, and the green line shows their three most familiar faces, all with statistical ranges shown as shaded areas. From Jayaraman *et al.* [34]

As Figure 6 illustrates, there's a simple but intriguing developmental pattern in infants' visual experience of faces. For very young babies, faces are visible approximately 15 minutes out of every hour, compared to only about 6 minutes per hour for older infants. In addition, for younger babies, these faces are typically very close (within 2 feet) and presented in frontal views. Interestingly, although infants need to learn many different faces throughout development, their early visual experience is concentrated on just a few familiar faces, providing massive exposure to these specific individuals.

However, when babies develop their motor skills—particularly when they start walking—the content of their visual experience changes significantly. Linda argues that such changes are important for how the human visual system develops, shifting from early experiences dense with faces to later experiences dense with hands manipulating objects. As Figure 7 shows, as babies grow, their exposure time to faces decreases significantly, a pattern observed in children from both the United States and India.

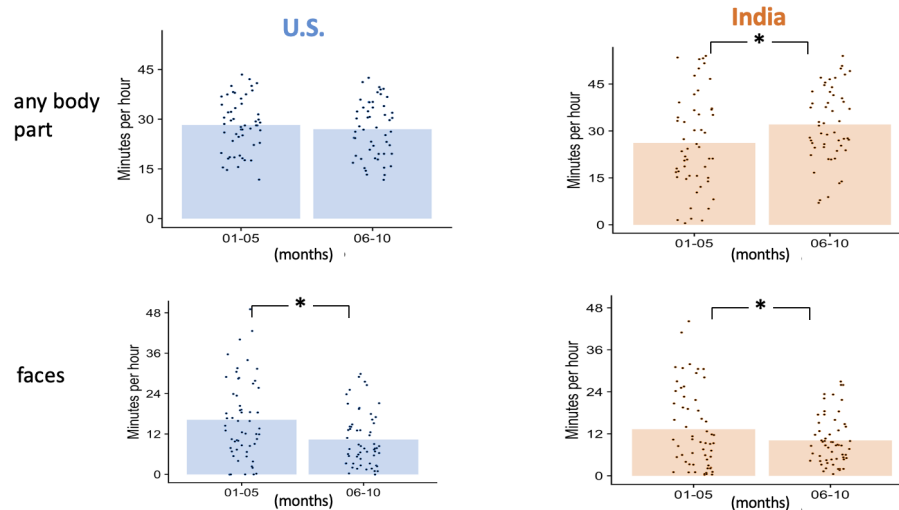


Figure 7: This figure shows how infants' visual experiences change with age across cultures. The top row shows exposure to any body part, while the bottom row shows exposure specifically to faces. In both the U.S. (blue) and India (orange), face exposure decreases significantly (*) as babies grow from 1-5 months to 6-10 months old. For body parts, exposure remains stable in U.S. infants but increases significantly in Indian infants with age. From Linda's slides.

To conclude this part, Linda argues that visual input heavily dominated by up-close faces and simple edges serves a crucial developmental purpose: it trains and tunes low-level feature extraction systems while simultaneously laying the foundation for higher semantic processing levels. Importantly, every visual experience—every moment of vision—carries information relevant to both low-level and high-level features. These processing levels develop together through bidirectional influences that are clearly evident during infancy, as demonstrated the paper [38]. Moreover, she made a point that sensory motor development creates different developmental niches for solving different visual learning problems, where for example, the contents of visual experience change: from dense with faces to dense with hands acting on objects. There are strong constraints on the data for learning within these niches, ensuring that infant environments contain the critical experiences. For instance, young babies need constant care, young babies like to look at what they can see, and parents like babies to look at them. These natural patterns create specialized visual input that optimally supports each stage of visual development.

2.3 Shape bias in early naming

In this part of the lecture, Linda introduces some circumstantial evidence of shape bias in early naming among infants. Please refer the original papers for more details [39–42].

Linda first introduces a series of laboratory studies that document children's evolving attention to shape in early word learning. In these controlled experiments, researchers present children with a novel object labeled with a novel name (e.g., "This is a dax"). When asked which other objects might share this same name, children's responses follow a systematic developmental progression: initially restricting the name solely to the specific demonstrated object, then gradually generalizing to objects sharing similar shapes. As shown in Figure 8, this shape bias strengthens predictably

with vocabulary growth between 14-20 months. This finding suggests that we can teach very young children to develop this generalizable shape bias earlier than it would naturally emerge, leading to faster vocabulary acquisition that persists up to a year later.

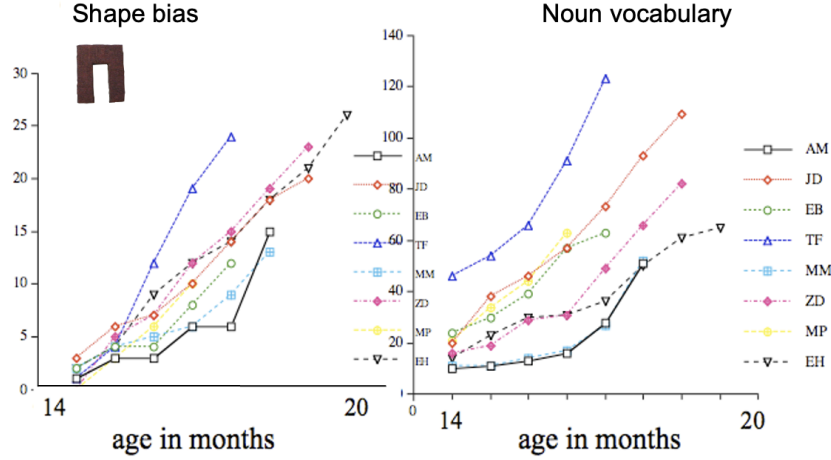


Figure 8: This figure illustrates the individual developmental relationship between shape bias and noun vocabulary growth in infants from 14 to 20 months of age. The left graph shows the increasing strength of shape bias (y-axis) as infants age (x-axis in months), while the right graph tracks noun vocabulary size (y-axis) over the same age range. From Gershkoff-Stowe & Smith [43].

2.4 What are the experiences that drive the developmental changes?

What kinds of visual experiences drive developmental changes in infants? One way to investigate this is by examining how toddlers' visual experiences differ from those of adults.

In the final part of this lecture, Linda presents a study from her team that quantifies the learnability of visual inputs using artificial neural networks. The data come from video recordings captured by first-person view cameras worn by both parents and children during naturalistic toy-play sessions [37]. The findings reveal that children's visual experiences are characterized by a distinctive combination of quality and diversity—featuring not only numerous large, high-quality views of objects but also a significantly broader range of rare perspectives—compared to those captured from the parents' viewpoint (see Figure 9).

What explains the differences between the visual experiences of parents and infants? Linda hypothesizes that a key factor is the temporal structure of the input. Unlike adults, infants do not process information in clearly segmented timescales; instead, they experience gradually evolving activation patterns across the visual field. As some studies suggest, this may reflect immature internal circuitry and ongoing neural development [44, 45]. However, these patterns

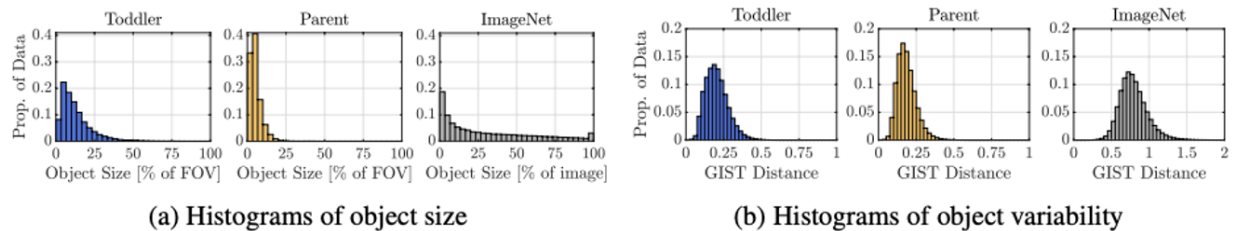


Figure 9: (a) Histograms of object size (as percentage of field of view) show that toddlers experience objects at larger sizes relative to their field of view compared to parents or standard ImageNet data. (b) Histograms of object variability (measured by GIST distance) demonstrate that toddlers experience both more consistent views (shown by the peak at smaller distances) and more diverse unusual views (shown by the longer tail) compared to parents. From Bambach *et al.* [2].

are also shaped by experience, suggesting that development is simultaneously constrained by biology and influenced by environmental input.

In her latest study, Linda and her team used video recordings from infants wearing head-mounted cameras to train various self-supervised learning models [3]. Critically, they separated the infant data by age group and evaluated the importance of training with a curriculum aligned with developmental order. The results showed that learning with data from the youngest age group provided the strongest learning signal and led to the best outcomes in terms of downstream task performance, as shown in Figure 10. Further analysis demonstrated that the benefits of the youngest age group’s data stem from the slowness and simplicity of their visual experiences. Please check the original paper for more details.

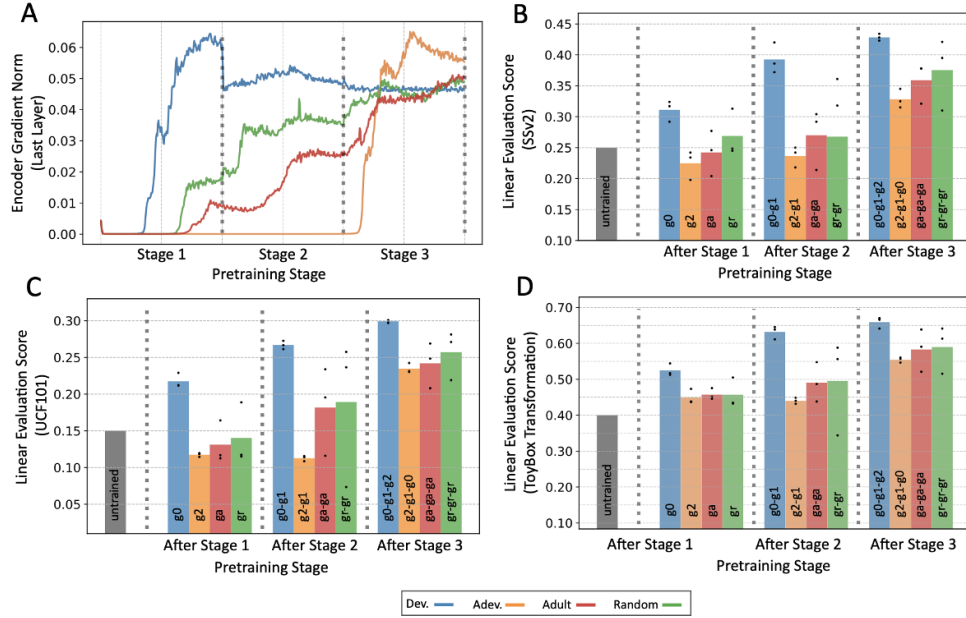


Figure 10: (A) shows gradient norms across three pretraining stages for different training sequences (developmental, anti-developmental, adult, and random). (B)-(D) display linear evaluation scores for three downstream tasks (SSv2, UCF101, and ToyBox transformation) measured after each training stage. The blue bars (Dev.) consistently show superior performance, suggesting that beginning training with data from the youngest age group provides the strongest learning signal and leads to better outcomes across all evaluation metrics. From Sheybani *et al.* [3].

2.5 Conclusion

In this lecture, Linda begins by highlighting how infants’ visual attention shifts developmentally in line with a natural curriculum—initially focusing on faces and gradually transitioning to hand-object interactions as they grow. She presents a compelling example of the shape bias in object naming, illustrating how perceptual categorization develops in tandem with language acquisition. The lecture concludes with empirical evidence from studies using artificial neural networks to quantify learning differences in visual data across developmental stages. These findings show that the visual experiences of younger infants offer uniquely valuable learning signals compared to those of older children or adults. Through these interconnected examples, Linda convincingly demonstrates that evolution, development, and real-time behavioral processes work in concert to create interdependence between internal learning mechanisms and the statistics of experience.

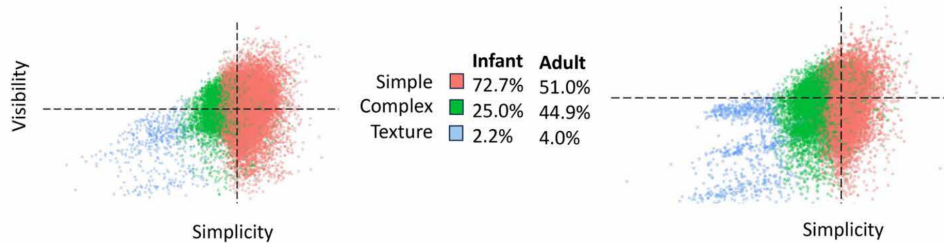


Figure 11: Distribution of infant (left) and adult (right) images in the simplicity-visibility space. The color of each dot indicates the cluster to which they were assigned in the cluster analysis. The table in the middle shows the proportions of the three clusters for both adults and infants. From Anderson *et al.* [15]

3 Lecture 3: Natural statistics

3.1 Introduction and preliminaries

Data is essential for learning processes in both human learners and artificial systems. As we witness unprecedented progress in machine learning capabilities, there emerges an urgent need to ground these learning processes in the physical world around us. To advance artificial agents toward human-level capacities in embodied learning, one crucial approach is to understand how statistical patterns in the natural world are organized and how they shape learning dynamics [46].

In this context, Linda argues that the physical world and human behavior share characteristic scale-invariant statistical properties that are fundamental for intelligence development. Human learning unfolds along a developmental trajectory with distinct sensitive periods that optimize information processing. For instance, early-stage visual development benefits from statistically biased inputs: faces appearing in close proximity provide high-contrast features, while simple edge patterns in less cluttered environments offer cleaner signal-to-noise ratios, all processed at slower rates that facilitate the sensitive period for vision. As development progresses, more complex statistical interactions emerge through hands manipulating objects, self-directed control of sensory input through bodily movements, and sustained attention mechanisms that collectively build sophisticated representations supporting object names, tool use, and higher cognitive functions.

In the following lecture, three key properties of natural statistics are introduced: skewed distributions, burstiness, and small-world networks. First, skewed distributions illustrate how information in nature is unevenly organized, with some patterns occurring much more frequently than others. Second, burstiness captures the temporal clustering of events, where related information tends to appear in short, concentrated bursts rather than being evenly spread over time. Third, small-world networks describe the highly interconnected structure through which information transitions in nature, combining local clustering with short global paths.

3.2 Property 1: Skewed distributions

The first property of natural statistics is the skewed distribution, often characterized by Zipf’s law. This statistical pattern appears consistently across diverse inputs that facilitate infant learning—whether analyzing individual faces, common object categories, or individual objects—all following a clear right-skewed rank order distribution.

Research by Linda and her team revealed these patterns through analysis of video recordings from head-mounted cameras worn by 8- to 10-month-old infants during 147 at-home mealtime events. Their findings uncovered a striking statistical signature: object category frequency distributions exhibited extreme right skew, with a very small set of objects appearing pervasively in infants’ visual fields, which may in turn reduce the challenge of referential ambiguity in early word learning. Their further analysis correlated these statistical properties with Age of Acquisition (AoA) data and demonstrated that the highest-frequency objects—those appearing more than 1,000 times across the recorded scenes—corresponded precisely with nouns in babies’ “First Words” acquisition, as shown in Figure 12. For complete details on this correlation, please refer to the original paper [47].

From these findings, Linda concluded that early experience is characterized by highly variable encounters with relatively few individual objects. Furthermore, controlled experimental studies strongly suggest that infant experiences with a very small number of unique instances (but many variable views of those things) is sufficient for broad, robust generalization [48–50]. In the final remark, Linda raised critical questions that remain unanswered in a world with

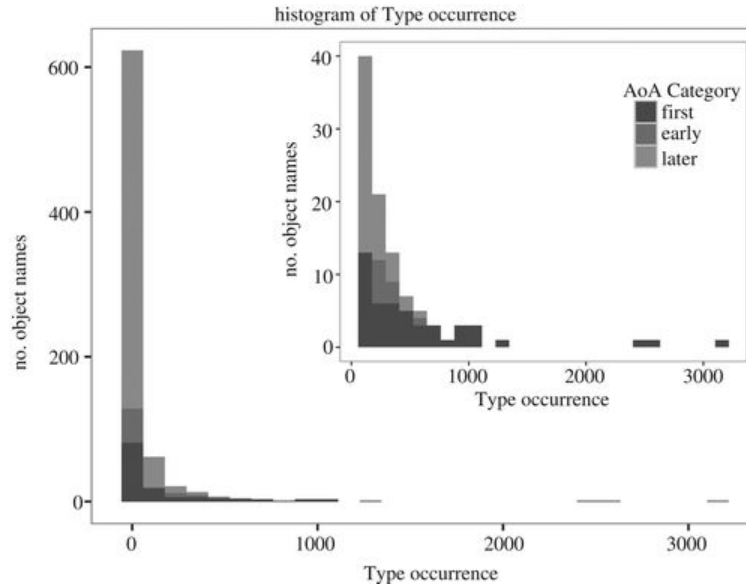


Figure 12: A histogram showing Type occurrence distribution across objects, with most items clustered near zero and few reaching 3000. The inset details this pattern by Age of Acquisition categories (first, early, later), illustrating a classic Zipfian distribution where few types occur frequently while most occur rarely. From Clerkin *et al.* [47].

continuity of space and time: What are the statistics of the variation in these highly frequent items? What is the role of the extensive “long tail”? What statistical principles govern these distributions?

3.3 Property 2: Burstiness

The second property of natural data is burstiness, a phenomenon where linguistic elements occur in temporal clusters rather than random distributions. This clustered pattern emerges because people typically talk about what they see and what they are doing—conversations that naturally follow contextual shifts rather than random patterns. This “bursty” context-dependent property of language has been documented across multiple timescales, from brief conversations to extensive texts [51–53]. Importantly, research consistently demonstrates that these temporal clusters benefit human learning [54–57].

To quantify temporal speech structure in natural contexts, Linda’s team analyzed parent-child interactions during free-play with toys. Their research reveals that within single play episodes, parent talk about individual objects occurs predominantly in temporal bursts—characterized by multiple references to the same object in quick succession, followed by periods with no mention of that object (as illustrated in Figure 13). For further details, see the original paper [58]

3.4 Property 3: Small world network

The third property of natural statistics, introduced by Linda, explores the collective dynamics of small-world networks, in which information is transmitted through highly clustered structures that maintain short path lengths despite having relatively few connections.

To empirically investigate such network structures, Linda and her team analyzed video recordings from head-mounted cameras worn by 1-year-old toddlers during controlled playroom experiments. In these sessions, toddlers interacted with their parents in a naturalistic environment where 33 toys were scattered across the floor. Parents were simply instructed to “play naturally,” allowing researchers to capture spontaneous, unstructured interactions. Each session lasted exactly 10 minutes—long enough to reveal interaction patterns while keeping the children’s attention engaged. For methodological details, see the original studies [59, 60].

To uncover the underlying network structure of toy play, the researchers applied graph-theoretic analyses to the time series of toy selections within each parent–child dyad. The results revealed that toy choices were not random or isolated; rather, they followed selective, recurring patterns centered around a small set of key toys, as illustrated in Figure 14.

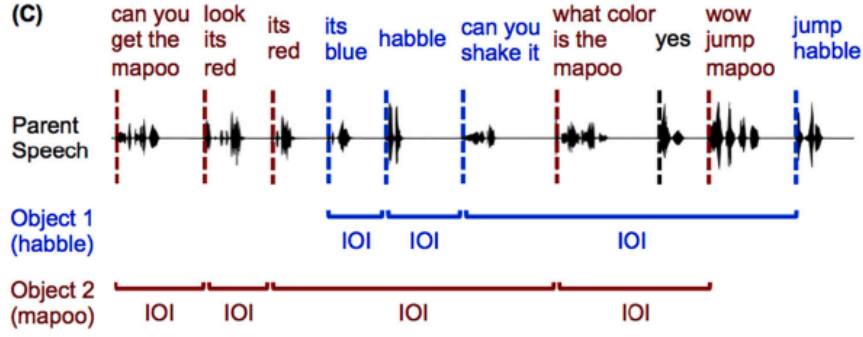


Figure 13: This figure shows the temporal structure of parent speech during object play. The waveform displays a parent’s utterances with text transcriptions color-coded to match reference objects: “habble” (blue) and “mapoo” (red). Below the speech waveform, two timelines represent moments when each object is referenced, with inter-onset intervals (IOIs) marked between consecutive mentions of the same object. The visualization demonstrates how parent speech about objects occurs in temporal bursts, with multiple references to the same object clustered together before switching to another object. From Slone *et al.* [58].

3.5 Conclusion

In this lecture, Linda introduces three key coherence statistics that underpin the structure of coherent and informative conversations and narratives: Zipfian distributions, burstiness, and network structure. She argues that the foundation of human generative intelligence lies in these statistical regularities, which emerge from behavior-generated experience. According to Linda, meaningful cognition arises from a system that produces coherent temporal patterns by connecting, repeating, and varying events. This system both learns from and generates new ideas through contextualized, time-extended experiences. Her perspective diverges from mainstream cognitive science by emphasizing that the relevant input for learning is not aggregated data, as in most machine learning approaches, but rather individual, story-like episodes of experience. Crucially, these inputs are co-constructed in real time through social interaction, with the learner actively shaping the temporal and co-occurrence patterns. The same internal mechanisms responsible for producing the stream of experience also support learning from it. Although these ideas remain empirically and computationally underdeveloped, they are deeply rooted in everyday human behavior—including that of toddlers—and in the memory processes that structure lived experience. Linda concludes that cognitive scientists should prioritize the study of time-extended coherence statistics in real-world episodes. Such behavior-generated experiences represent both the

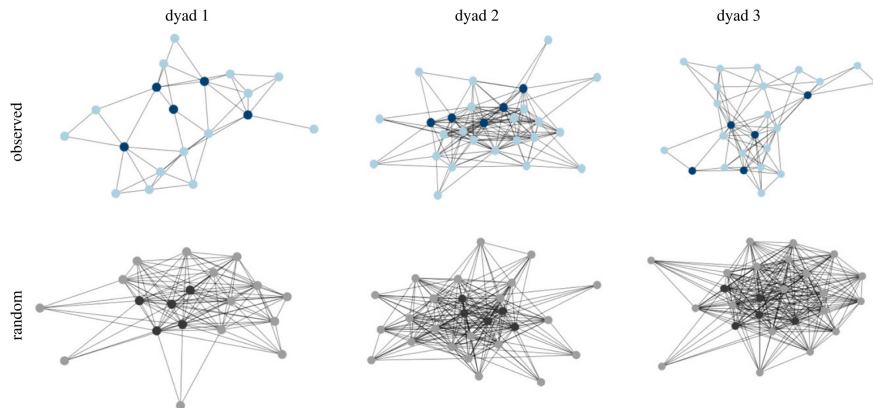


Figure 14: The figure shows the network structures across three parent-child dyads, contrasting observed toy transition patterns (top row, blue nodes) with random permutation models (bottom row, gray nodes). Each network visualizes how interactions flow between different toys, with nodes representing toys and connections showing transitions between them. The observed networks display more structured, less densely connected patterns than their random counterparts. From Karmazyn-Raz & Smith [59].

evolutionary and developmental context of human intelligence, and the kind of environment in which AI systems may ultimately operate.

4 Lecture 4: The dynamic coupling of parents and toddlers

4.1 Introduction and preliminaries

Parent–toddler interactions are complex systems with multiple pathways of engagement. The relationship between parents and toddlers involves dynamic coupling that operates through various routes, all of which are important for development. It is known that children show preferences for different toys, rather than all gravitating toward the same popular items. However, some consistent patterns emerged across all parent–child dyads: each pair tended to play extensively with a small subset of toys while interacting only briefly with the majority of available ones. There were strong correlations in behavior between parents and children directed toward specific objects. These correlations were evident not only in the frequency of interaction with particular toys but also in the timing of engagement.

When a parent interacts with their child, a “magic moment” for learning can occur—two distinct minds temporarily aligning. This lecture illustrates how such moments emerge from the dynamic coupling of parent–toddler behavior, and how they foster early word learning and human development.

To investigate these dynamics, Linda and her team has spent nearly a decade assembling a uniquely detailed corpus, referred to as the Multi-Sensory Project—a large, temporally dense dataset from over 200 parent–child dyads (ages 9–30 months) [7, 20, 26, 61]. Each parent–infant dyad wears dual head-mounted eye trackers (or head cameras), motion sensors on heads and hands, and is recorded by multiple room cameras plus high-fidelity audio in both lab and home settings. This setup captures temporally dense records of what each partner sees, touches, and says, allowing analyses to probe the “magic moment” at the sensorimotor level—how the coordination of gaze, hand, and speech gives rise to shared attention and, ultimately, to word learning.

This lecture will outline four key analytical results from the dataset, demonstrating that the dynamic coupling between parent and child is not simply a magical transmitter of knowledge. While the coupling matters, it’s ultimately the child who does the learning, and the child’s own dynamics play a crucial role in this process.

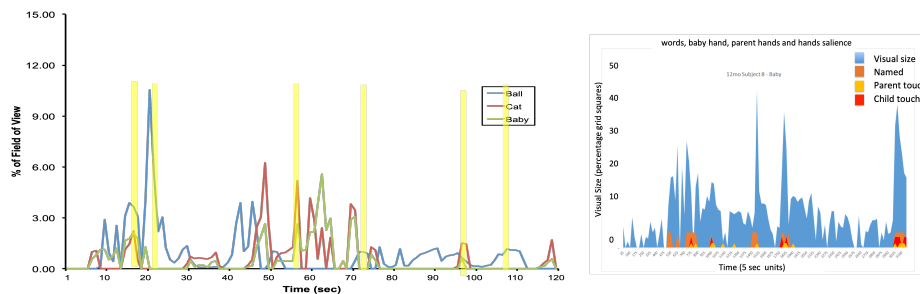


Figure 15: Left: This figure shows when during play the mother—who was interacting and playing with her infant—named the object, where parents often name objects at moments in which one object dominates. Right: This figure demonstrates that this pattern characterizes visual experience and object naming at home and in more cluttered contexts. Both figures are adapted from Linda’s lecture slides.

4.2 Part 1. Child’s perspective

This first half of the lecture zooms in on the sensory-motor dynamics from **child’s perspective** of early object naming learning. The discussion centers on methodologies and findings from a few papers published by Linda’s research team [7, 19, 20, 61]. For comprehensive details, please refer to the original publications.

The goal of the studies is to compare low-competition scenes—where one object occupies the toddler’s view—to high-competition ones with multiple nameable objects jostling for attention. To investigate this, 18-month-old toddlers wore lightweight head-mounted eye trackers, allowing researchers to capture statistics that reflect what the baby actually sees, touches, and hears in real time. Here are the two main takeaways from the studies.

1. Bodily “Zoom-In”: Brief, High-Salience Windows

Toddlers in natural play often *create* their own highly focused windows by moving the object closer to their eyes or by leaning their torso and head toward it. From a toddler’s perspective, these convergences of motion and naming are special moments, and they coincide with the parent’s label to create an optimal moment for learning object names.

2. Parent Naming Overlaps with Child’s Salience Peaks

Parents time their labels to the child’s high-salience moments in which one object dominates rather than periods of visual competition. As shown in Figure 15 (Left), almost every naming token coincides with a steep peak in the focal object’s size curve—exactly those transient windows when one toy dominates the child’s field of view. The same one-object-dominates principle holds outside the lab (as shown in Figure 15 (right)), even in messy, toy-strewn living rooms. A 12-month-old’s head-camera video at 5 Hz was used to compute the pixel footprint of a single baby-doll across a 12×18 grid (216 squares). The time-series (blue fill) again shows sharp spikes whenever the doll swells in view, and the parent’s naming tokens cluster under those spikes.

From the toddler’s perspective, these moments are visually special and they coincide with naming by the parent. A natural question is whether these moments become optimal for learning object names. To test whether those visually salient naming moments actually produce vocabulary growth, Linda and her team ran a play–test protocol. Parents first played with six novel objects, each paired with a nonsense label (e.g., zeebee, tema, dodi). Immediately afterward, toddlers completed a two-alternative forced-choice task; a word was classified as learned if the child selected the correct referent on both of two trials (2 / 2), and unlearned otherwise. They then compared the perceptual profile of the naming episodes that preceded learned versus unlearned words [19, 20] as shown in Figure 16.

Their studies show that multimodal salience is a key factor in word learning: children are more likely to learn words when the named object is larger, more centered, and more visually stable than competing objects. Learning is enhanced when the target object clearly dominates the visual field. Moreover, the duration and continuity of these moments matter—toddlers need several seconds of an uncrowded, high-salience interval to successfully bind a new label to its referent.

In summary, visual attention in toddlers involves more than just eye movements—it’s enacted through the coordinated self-organization of head, eyes, and hands. When these body parts converge on a single object, visual competition collapses for several seconds, creating stability in the visual field. During successful naming moments, visual properties such as size, centeredness, and brightness reliably rise together. Importantly, naming events with these stacked, enduring properties strongly predict which novel words toddlers remember. This coupling of eyes, heads, and hands supports learning by stabilizing the child’s visual field and reducing visual competition. By creating these periods of visual stability, toddlers can more effectively process and learn new words in their environment.

4.3 Part 2. Parent’s perspective

So what about the mature partner’s side? What does the mature partner know about these optimal moments for learning object names? Figure 17 juxtaposes the size-over-time traces for the same three toys as captured by the child’s head



Figure 16: This figure illustrates a word learning experiment 1) Novel Words & Objects (left) showing six colorful toy shapes paired with novel words (“zeebee,” “tema,” etc.); 2) Parent-Toddler Free Play (center) depicting a parent and child with head-mounted eye-tracking equipment engaged in play with the novel objects; and 3) Word Learning Test (right) showing a child being assessed on word-object association. From Pereira *et al.* [19].

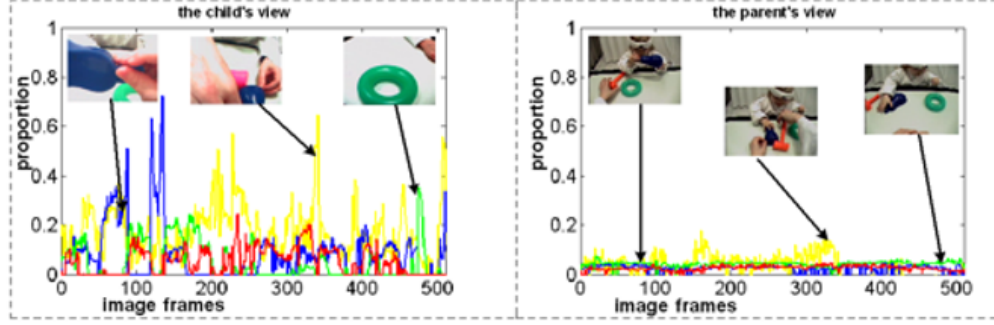


Figure 17: The figure displays the size of objects (measured as proportion of pixels) in two simultaneous visual perspectives: the parent’s view and the child’s view, plotted as a function of time. From Smith *et al.*

camera (left) and the parent’s head camera (right). From the toddler’s vantage point, the plot is dominated by tall, spiky peaks: one object regularly balloons to occupy most of the pixel grid, while competitors plummet toward zero. By contrast, the parent’s view presents a flat landscape—no object ever dominates the majority of the frame. In other words, the parent almost never witnesses the visually “magic” moment that the child experiences.

As a result, parents use indirect signals—e.g., the child’s hand and head orientation—to infer “now is a good time” to name the object. They cannot see the toddler’s *actual* low-competition view but rely on inference (where the toy is, who is holding it, whether it is moving). For moment-to-moment with the parent, the visual size of the object does not change for the adult’s own view. For this reason, the parent uses gaze shifts or hand movements to guide the child, rather than “matching” the toddler’s actual visual dominance.

A critical question emerges: what predicts when parents choose to name objects, and what predicts when infants successfully learn those names? Are these processes governed by the same factors? In the follow-up study [9], the parent’s decision rule was probed in a head-camera study of 12 dyads (18- to 22 months) playing with nine novel objects/labels. After the free-play phase, toddlers were tested on those labels (Figure 18); this design parallels the previous part, allowing comparison of which cues predict *learning* versus which cues trigger *parental naming*.

Figure 19 shows that the five measures are not independent: objects that are visually large are often also near centre, and “held-by-parent” co-occurs with motion because the adult tends to pivot or rotate the toy while naming. The largest gaps between naming (dark bars) and no-naming (light bars) appear in object motion and parent holding—suggesting that parents are especially likely to speak when the referent is in their own hand or is moving (spinning, rolling) regardless of who moves it. Together these patterns reinforce the idea that caregivers exploit a bundle of partially redundant cues—size, centredness, motion, and hand contact—to time their labels.

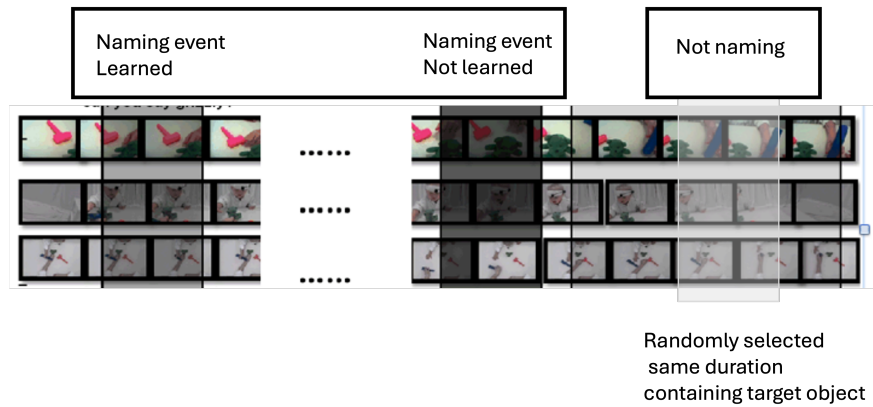


Figure 18: This figure shows the annotation scheme: **Naming + learned**: The parent utters the label, and the child later succeeds on the test for that word. **Naming + unlearned**: The parent utters the label, but the child does not retain it; **No-naming controls**: Equal-duration windows in which the same target object is visible but is *not* labeled. From Suanda *et al.* [9].

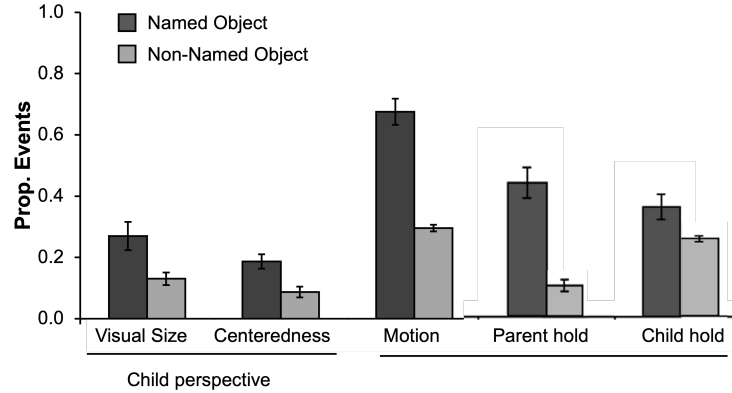


Figure 19: The figure shows the proportions of occurrences of naming object events versus non-naming object events with respect to five measures. Five measures were extracted for all objects (the target and competitors): a) Visual size in the toddler’s view. b) Distance to center in the toddler’s view. c) Motion of the object in the scene. d) Held by the toddler. e) Held by the parent. From Smith *et al.* [61].

The further statistical analysis reveals that parents pay attention to centeredness, motion, and who holds the object. The final pathway to successful word mapping is a set of visual properties available only in the child’s own view: dominance, stillness, and sustained centredness. Parents cannot see these states directly; at best they infer them from body geometry and ongoing motion—and sometimes those inferences miss the mark. Motion, for instance, attracts the adult’s urge to speak yet undermines the visual stabilization the learner needs. In short, the dyad works because the caregiver’s reasonable but imperfect hypotheses about “good naming moments” intersect—just often enough—with the actual perceptual conditions that drive learning.

To conclude part 2, parents aren’t mind-readers when identifying optimal moments for teaching vocabulary. What they are trying to do is challenging rather than magical; they must rely on observable cues without direct access to their baby’s visual perspective. Parents use perceptible signals like head orientation and hand movements to infer appropriate naming opportunities, employing a best-guess strategy built on partial, body-based information. This approach yields good but imperfect results, with approximately 2.5 “sub-optimal” naming events occurring for every “optimal” one. In real-world scenarios, some parents demonstrate greater skill at this perceptual inference than others. However, the underlying mechanism remains fundamentally non-magical across all parent-child interactions—it’s simply a strategic inferential process working with limited information.

4.4 Part 3: Is there magic in the complex dynamic coupling of bodies?

In this section, Linda introduced a series of experiments involving dual head-mounted eye-tracking, to investigate the complex dynamics coupling of parent-child behavior. For detailed methods and results, please refer to the original papers [62–64].

Their studies reveal distinctly different gaze patterns between children and parents during interactions. Child gaze can be characterized as “sticky”—they dwell on objects substantially longer than adults do, maintaining sustained attention on individual items. In contrast, parent gaze is “flitting”—they rapidly sample among toys and faces to efficiently monitor and scaffold play activities. Interestingly, toddlers rarely shift attention to their mother’s face when handling objects, preferring to maintain visual focus on the items themselves. Despite these contrasting visual strategies, parent-child dyads achieve sustained joint attention on the same object more than 50% of the time during play interactions. This successful coordination occurs not because parents and children use identical looking patterns, but because parents adapt their gaze behavior to accommodate and support their child’s more focused visual engagement.

Figure 20 shows that toddlers look at the parent’s face only ~6% of the time, yet manage to align their attention with the parent on the same object more than 50% of the time. Hence, joint attention in free play does not rely on explicit face-to-face gaze following.

Figure 21 shows that the left side of the curve (negative lags) represents episodes in which the child leads and the parent joins; the right side represents parent leads. The profile is strikingly symmetric, but the modal lag for child-lead bouts is shorter than for parent-lead bouts, indicating that infants are, on average, quicker to follow. The lag-based

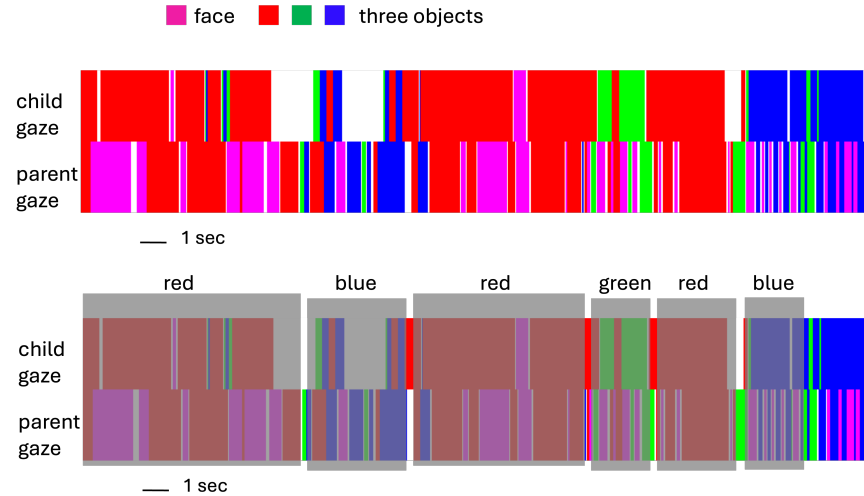


Figure 20: The figure presents a detailed time series capturing the moment-by-moment fixation patterns of one parent-child dyad during natural play. Adapted from the Linda's slides.

metrics reinforce the picture of a dynamically coupled system in which either partner can initiate alignment and the other can lock on rapidly—akin to dancing partners.

How about the eye-hand coordination? Within each individual, eyes and hands are tightly coupled. Across partners, following one another's manual actions offers a direct, efficient route into shared object focus—eliminating the need for explicit face or gaze reading. This eye-hand linkage helps both parent and child converge on the same target, further strengthening their coordinated engagement.

In summary, when parent and child achieve close coordination, three forces converge to stabilize the child's visual experience of the target object. First, head stability significantly increases, with fewer positional and orientational shifts occurring, which effectively reduces visual jitter in the child's field of view. Second, holding behavior becomes more consistent, as the child (and frequently the parent) firmly maintains grasp on the same object, minimizing sudden changes in position that would otherwise disrupt visual attention. Third, eye-head alignment improves substantially, with the toddler's gaze remaining locked near the head-camera center, creating optimal conditions for sustained visual

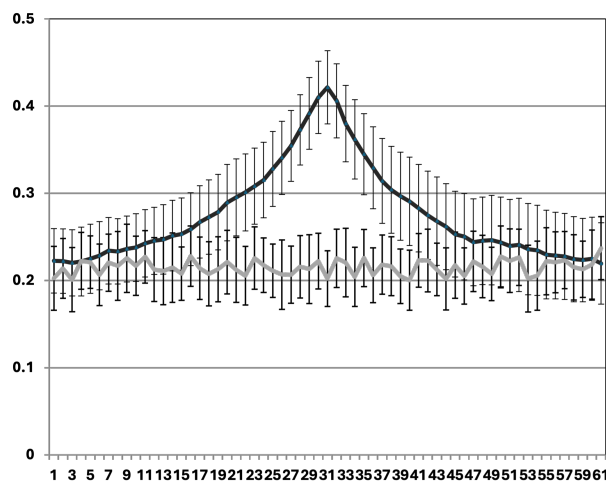


Figure 21: The figure presents a diagonal-wise recurrence lag profile that represents the pattern of coordination between the two gaze streams, adapted from Linda's slides.

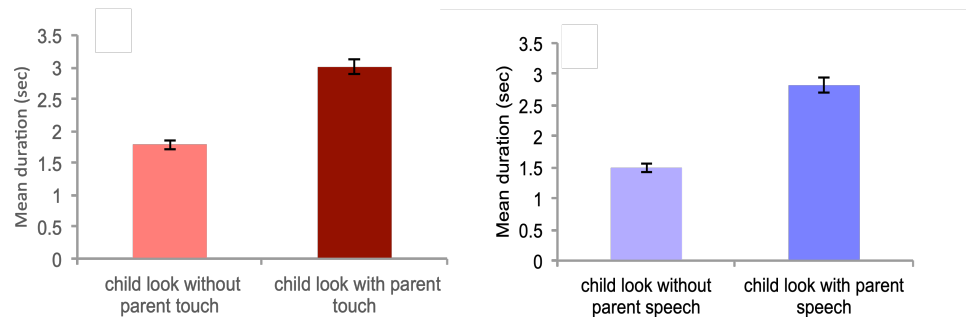


Figure 22: The figure shows the average duration of the child's gaze at the target, comparing conditions with and without parent touch, and with and without parent speech.

focus. Together, these factors create a stable visual window for the infant, supporting enduring visual dominance of attention to the target object.

4.5 Part 4: What does coupled parent and child hands and eyes have to do with the stable child head, hands and eyes, all focused on a single object?

In the final segment of the lecture, Linda shifts focus to address the central question: What implications do these findings have for object learning, and how does joint attention matter for development? Traditionally, the ability to sustain attention has been conceptualized as an individual property of the infant—a developmental product of increasing self-regulatory and endogenous control (i.e., internal, top-down, voluntary) over one's attention and cognitive systems [65]. For example Ruff & Rothbart define a threshold of three seconds of consecutive attention to be “sustained” [65]. However, this perspective may be incomplete. Recent research using head-mounted eye trackers during object play has provided a more nuanced view of attention patterns. The data reveals a right-skewed distribution of attention durations, with many brief attentional episodes but also some remarkably extended ones. Recent measurements from 12-month-olds indicate that approximately 19% of those infants' object-focused attention during play extends beyond 3 seconds, with some episodes approaching 8 seconds in duration [8].

Linda then examines instances of sustained attention (SA) in relation to joint attention (JA). Using a median split, she categorizes these episodes into long sustained attention and short sustained attention. For each SA instance, she analyzes the accompanying joint attention periods, dividing sustained attention into three temporal segments: before joint attention, during joint attention, and after joint attention. The analysis reveals a significant relationship between these attention types. Sustained attention bouts that occur with joint attention (SA+JA) last notably longer—approximately 5.3 seconds versus approximately 4.4 seconds without joint attention. Furthermore, the data demonstrates a clear dose-dependent extension effect: the longer the parent maintains visual attention on the same object, the more persistently the child's attention is sustained.

Longer JA segments predict longer total SA, and the after phase is extended in proportion to the preceding JA length—indicating a lasting, dose-dependent boost from parental co-attention. The time it takes the parent to join the child is not related to how long the child has already been looking. But once the parent does join, the sustained attention extends further.

The analysis further reveals that longer joint attention segments predict longer total sustained attention periods. Importantly, the “after phase”—when the child continues looking at an object after the parent has disengaged—is extended in proportion to the preceding joint attention duration. This pattern indicates a lasting, dose-dependent boost from parental co-attention that persists beyond the immediate interaction. Interestingly, the time it takes for a parent to join the child's attentional focus is not related to how long the child has already been looking at the object. This suggests that children's initial engagement with objects occurs independently of parental involvement. However, once the parent does join the child's attentional focus, the child's sustained attention extends significantly further.

So toddlers rarely look at their parents' faces during object exploration, raising the question: How do they “know” when parents are also attending to the same object? Linda argues that during physical engagement with objects, each person's body forms a coupled system in which eyes, head, and hands work in coordinated harmony. Figure 22 illustrates two critical findings: when the parent physically touches the object the child is already fixating on, the child's attention bouts significantly increase in duration; similarly, when the parent speaks while the infant is looking at an object, the duration of the child's visual attention jumps considerably. Furthermore, the latest studies from Linda's

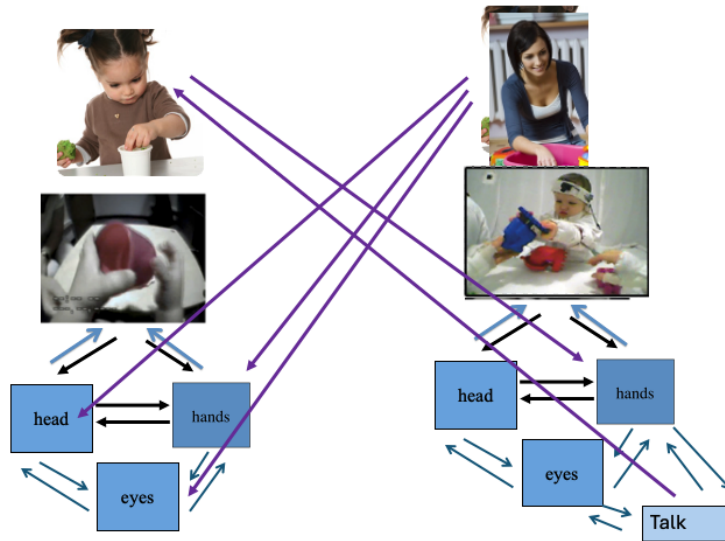


Figure 23: The figure illustrates the idea of through the Body – Indirect Brain-to-Brain Coupling, which is taken from Linda’s slides.

team confirm that even in messy, everyday settings, parents name objects when the referent visibly dominates the infant’s view—and they often touch that object while naming. Thus, the “magic moments” in the lab generalize to natural home play.

In summary, each partner in the interaction functions as a self-contained sensorimotor system, creating a situation where “brain-to-brain” coupling can only occur through bodily actions on shared objects. Information flows sequentially: from brain to head, eyes, and hands, then to the object, and ultimately to the partner’s own hands, eyes, head, and brain. As Figure 23 illustrates, parents’ verbal commentary and physical touches highlight the joint referent, thereby extending the child’s sustained attention. Together, parent and child form a flexible mesh in which each body part can rapidly drive or adjust to the other’s state. Recent research further suggests that mother and child, while operating as distinct systems, coordinate and follow each other’s bodily actions in ways that yield a form of brain-to-brain coupling via autonomic system attention mechanisms (Wass, 2022; Endevelt-Shapira, 2023). From an “extended brain” perspective, the brain extends into the world through the body—and any coupling between brains necessarily flows from one brain through its body to the shared world, and then through the partner’s body to their brain. This embodied view of social cognition helps explain how learning emerges from physically grounded interactions rather than abstract mental connections.

4.6 Conclusion

This lecture highlights that learning does not occur through the direct transfer of knowledge from parent to child. Rather than functioning as a conduit through which expertise is passed wholesale from expert to novice, the dynamic coupling between parent and child supports learning that ultimately unfolds within the child’s own embodied experience, grounded in the infant’s sensorimotor activity.: it is the infant’s sensorimotor dynamics—the coordinated trajectories of eyes, head, and hands—that stabilize perception and support successful word mapping. This evidence-based reframing challenges us to look beyond the popular narrative of “joint attention as information transfer.” Instead, it directs us toward a systems view in which a caregiver and child collaboratively create the optimal input stream from which the child can most readily learn. Rather than conceptualizing learning as knowledge passing directly from one mind to another, we should understand it as an emergent property of physically grounded interactions that create stable perceptual experiences. This perspective illuminates why certain moments of interaction are particularly powerful for language acquisition and provides a more mechanistic account of how social engagement supports cognitive development.

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